MILITARY SPECIFICATION

INVERTER, POWER, STATIC: 10516412

This amendment forms a part of Military Specification MIL-I-60166(MU), dated 14 May 1964.

Paragraph 2.1: Delete the following Military Specification:

"MIL-I-26600 Interference Control Requirements, Aeronautical Equipment"

and substitute the following Military Standards:

"MIL-STD-461 Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of"

* Paragraph 3.1: Delete paragraph.

* Paragraph 3.4: Delete "600 in a horizontal position and 400 hours in a position 180° from horizontal" and substitute "60 hours in a horizontal position and 40 hours in a position 180° from horizontal".

Paragraph 3.5.2: Delete last sentence and substitute:

"The inverter shall meet the requirements of 3.7 before and after vibration."

Paragraph 3.5.2.1: Add the following new paragraph:

"Monitoring during vibration.- The inverter input voltage shall be 24 vdc. A load of 150 volt amperes at unit power factor shall be applied. The characteristics to be monitored are as follows: input current, voltage output, output current, frequency of output voltage, and the harmonic content of the output waveform. All monitored values shall meet the requirements of 3.7."

Paragraph 3.7: Delete first sentence and substitute the following:

"Before and after vibration (see 3.5.2) the inverter shall meet the requirements of 3.7.1 thru 3.7.6 without spasmodic or intermittent operation, with input voltages of 18 vdc, 24 vdc and 30 vdc."

Paragraph 3.7.5: Delete and substitute the following:

"3.7.5 Harmonic distortion.- The total peak output voltage of the inverter shall not exceed 1.25 times the peak to peak output voltage specified in

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3.7.2 and 3.7.3. Duration of transients at the output shall diminish to 1.10 times the peak to peak output voltage, or less, within three cycles. The above requirement shall apply during energizing and deenergizing the inverter under all conditions of output load. The inverter shall have a harmonic distortion of less than 4% for all loads and all power factor ranges between 0.8 leading and 0.6 lagging. With a unity power factor and at a load of 75 VA, the maximum variation of harmonic distortion shall be 1.5%.

Paragraph 3.10: Delete and substitute the following:

3.10 Radio Interference:- The inverter shall meet the following requirements of MIL-STD-461, Class IC; CE03 (20 KHz to 50 MHz), CS01 (3KHz to 50 KHz), RE02 (14KHz to 1 GHz) (T) RE04 (20 KHz to 50 KHz). Tests are to be performed for broadband interference only.

3.11 Continuity.- The outside case shall be constructed in such a manner that it will provide one conductance path between any two points on the case."

* Following paragraph 4.1.1, add paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2.

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) inverters manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Paragraphs 4.2, 4.2.1, 4.2.1.1, 4.2.1.2: Delete in their entirety.

Paragraph 4.3.2, Table II: Delete the following:

"113. Radio interference	3.10	4.6.11"

and substitute:

"115. Continuity	3.11	4.6.12"

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Paragraph 4.4.1: Add the following control test:

"Radio interference 3.10 4.6.11"

Paragraph 4.4.3, Reliability: Delete in its entirety.

* Paragraph 4.6.5: Add the following sentence:

"This test shall only be conducted for First Article (see 4.1.2).

Paragraph 4.6.6.2: Delete the second sentence and substitute the following:

"Before and after vibration, the inverter shall be tested at inputs 18 vdc, 24 vdc and 30 vdc in accordance with 4.6.8 to 4.6.8.6, inclusive to determine compliance with 3.7. During vibration in each of the three mutually perpendicular planes (see 3.5.2), the inverter shall be monitored when energized at 24 vdc with a unity power factor load of 150 VA to determine compliance with 3.5.2.1."

Paragraph 4.6.8.5: Delete last sentence and add the following:

"4.6.8.5 Harmonic distortion.- All measurements shall be made with standard measuring equipment, utilizing a Tektronic #564 storage oscilliscope with a No. 3A1 plug-in module or equivalent test equipment for measuring voltage spikes during switching between loads as specified in 3.7.5."

Paragraph 4.6.8.6, third sentence: Delete "for a minimum of one hour" and substitute "and be stabilized for a period of five minutes."

Paragraph 4.6.11: Delete "MIL-I-26600" and substitute "MIL-STD-462".

Add paragraph 4.6.12 Continuity.- The test for continuity shall be conducted as an inprocess test prior to application of paint finish on the exterior surfaces, to insure that the case mating surfaces have metal to metal contact. This test shall be conducted by placing a potential ground. Using standard measuring equipment, measure the potential between the other five sides and ground potential. Measured values of this potential should be at least 90 ± 5 milli-volts to insure proper continuity as specified in 3.11.

* Paragraph 6.3: Delete paragraph.

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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